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A four-component hierarchy of combined integrable equations with bi-Hamiltonian formulations

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ABSTRACT

Upon introducing a specific 4×4 matrix eigenvalue problem with four components, we would like to construct a Liouville integrable Hamiltonian hierarchy, within the zero curvature formulation. Bi-Hamiltonian formulations are furnished via the trace identity, through which the Liouville integrability of the resulting hierarchy is explored. The first two nonlinear examples are novel generalized combined nonlinear Schrödinger equations and modified Korteweg–de Vries equations.

1. Introduction

Integrable models can be generated from Lax pairs of matrix eigenvalue problems [1], based on which one can explore remarkable integrable properties, for example, inverse scattering transforms and bi-Hamiltonian formulations [2,3]. There are abundant applications of integrable models in physical and engineering sciences, including nonlinear optics, plasma physics, water waves and quantum mechanics.

Among typical integrable models are the Ablowitz-Kaup–Newell–Segur hierarchy [4] and its various hierarchies of integrable couplings [5]. Matrix loop algebras are the starting point of formulating meaningful Lax pairs [6,7]. In this paper, we would like to propose a new specific matrix eigenvalue problem and generate an associated integrable hierarchy.

The steps of the zero curvature formulation to construct integrable hierarchies are as follows (see [6,7] for details). First, based on a given loop matrix algebra $\tilde{g}$ with the loop parameter λ , we formulate a spatial spectral matrix:

$$\mathcal{M} = \mathcal{M}(u, \lambda) = u_1 E_1(\lambda) + \cdots + u_q E_q(\lambda) + E_0(\lambda), \quad (1.1)$$

where $u = (u_1, \dots, u_q)^T$, the elements $E_1, \dots, E_q$ are linear independent in $\tilde{g}$, and the element E_0 satisfies the pseudo-regular conditions: $\text{Im ad}_{E_0} \oplus \text{Ker ad}_{E_0} = \tilde{g}$, $[\text{Ker ad}_{E_0}, \text{Ker ad}_{E_0}] = 0$, in which ad_{E_0} stands for the adjoint action of E_0 on $\tilde{g}$. These pseudo-regular conditions guarantee that there is a Laurent series solution $Y = \sum_{n \geq 0} \lambda^{-n} Y^{[n]}$ in $\tilde{g}$ to the stationary zero curvature equation

$$Y_x = [\mathcal{M}, Y]. \quad (1.2)$$

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Second, we introduce an infinite sequence of temporal spectral matrices

$$\mathcal{N}^{[m]} = \mathcal{N}^{[m]}(u, \lambda) = (\lambda^m Y)_+ + \Delta_r = \sum_{n=0}^m \lambda^{m-n} Y^{[n]} + \Delta_m, \quad m \geq 0, \quad (1.3)$$

where $\Delta_m \in \tilde{g}$, $m \geq 0$, so that a hierarchy of integrable models:

$$u_{t_m} = X^{[m]} = X^{[m]}(u), \quad m \geq 0, \quad (1.4)$$

can be generated from the zero curvature equations:

$$\mathcal{M}_{t_m} - \mathcal{N}_x^{[m]} + [\mathcal{M}, \mathcal{N}^{[m]}] = 0, \quad m \geq 0, \quad (1.5)$$

which are the solvability conditions of the spatial and temporal matrix eigenvalue problems:

$$\varphi_x = \mathcal{M}(u, \lambda)\varphi, \quad \varphi_{t_m} = \mathcal{N}^{[m]}(u, \lambda)\varphi, \quad m \geq 0. \quad (1.6)$$

Third, taking use of the so-called trace identity:

$$\frac{\delta}{\delta u} \int \text{tr} \left(Y \frac{\partial \mathcal{M}}{\partial \lambda} \right) dx = \lambda^{-\kappa} \frac{\partial}{\partial \lambda} \lambda^{\kappa} \text{tr} \left(Y \frac{\partial \mathcal{M}}{\partial u} \right), \quad (1.7)$$

where κ is a constant, we furnish the equations in the hierarchy (1.4) with bi-Hamiltonian formulations and thus show their Liouville integrability.

Many hierarchies of Liouville integrable models have been explored [4–17]. The noteworthy integrable hierarchies with two components include the Ablowitz–Kaup–Newell–Segur hierarchy [4], the Heisenberg hierarchy [18], the Kaup–Newell hierarchy [19] and the Wadati–Konno–Ichikawa hierarchy [20].

The aim of this paper is to present an addition to integrable hierarchies. Upon introducing a specific eigenvalue problem, we would like to construct a four-component integrable hierarchy through the zero curvature formulation. The corresponding bi-Hamiltonian formulations are established via the so-called trace identity. Two concrete examples are novel generalized combined integrable nonlinear Schrödinger and modified Korteweg–de Vries models. The last section presents a conclusion, together with concluding remarks.

2. Lax pairs and a four-component soliton hierarchy

Let δ be an arbitrary constant, and T be an r th-order square matrix such that $T^2 = -I_r$, where I_r denotes the r th-order identity matrix. A set of block matrices

$$g = \left\{ A = \begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix}_{2r \times 2r} \mid A_4 = T A_1 T^{-1}, \quad A_3 = \delta T A_2 T^{-1} \right\} \quad (2.1)$$

forms a matrix Lie algebra under the matrix commutator $[A, B] = AB - BA$. We will use the Lie algebra with $r = 2$ and

$$T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad (2.2)$$

to formulate a specific spectral matrix below.

Let α_1 and α_2 be two numbers, and $u = u(x, t) = (u_1, u_2, u_3, u_4)^T$ be a four-component potential vector. Assume that

$$\alpha = \alpha_1 - \alpha_2 \neq 0. \quad (2.3)$$

Motivated by recent studies on integrable hierarchies with four potentials (see, e.g., [21–25]), we introduce a 4×4 matrix eigenvalue problem of the form:

$$\varphi_x = \mathcal{M}\varphi = \mathcal{M}(u, \lambda)\varphi, \quad \mathcal{M} = \begin{bmatrix} \alpha_1 \lambda & \delta_1 u_1 & u_2 & 0 \\ \delta_1 u_3 & \alpha_2 \lambda & 0 & u_4 \\ \delta_1 \delta_2 u_4 & 0 & \alpha_2 \lambda & -\delta_1 u_3 \\ 0 & \delta_1 \delta_2 u_2 & -\delta_1 u_1 & \alpha_1 \lambda \end{bmatrix}, \quad (2.4)$$

where δ_1 and δ_2 are two arbitrary constants. This spectral matrix $\mathcal{M}$ belongs to a loop algebra associated with the previous matrix algebra g , with $r = 2$ and T being defined by (2.2). The eigenvalue problem cannot be reduced from the matrix Ablowitz–Kaup–Newell–Segur one (see, e.g., [26]). From this eigenvalue problem, we will generate an integrable hierarchy of bi-Hamiltonian equations, which possess particular combined structures. Interestingly, the case of $\delta_2 = 0$ reduces to integrable couplings, which are not of perturbation type.

We first solve the corresponding stationary zero curvature Eq. (1.2) by assuming Y to be of the following Laurent series type:

$$Y = \begin{bmatrix} \delta_1 a & \delta_1 b & e & f \\ \delta_1 c & -\delta_1 a & f & g \\ \delta_1 \delta_2 g & -\delta_1 \delta_2 f & -\delta_1 a & -\delta_1 c \\ -\delta_1 \delta_2 f & \delta_1 \delta_2 e & -\delta_1 b & \delta_1 a \end{bmatrix} = \sum_{n \geq 0} \lambda^{-n} Y^{[n]}, \quad (2.5)$$

where the basic objects can be stated as follows:

$$\begin{cases} a = \sum_{n \geq 0} \lambda^{-n} a^{[n]}, & b = \sum_{n \geq 0} \lambda^{-n} b^{[n]}, & c = \sum_{n \geq 0} \lambda^{-n} c^{[n]}, \\ e = \sum_{n \geq 0} \lambda^{-n} e^{[n]}, & f = \sum_{n \geq 0} \lambda^{-n} f^{[n]}, & g = \sum_{n \geq 0} \lambda^{-n} g^{[n]}. \end{cases} \quad (2.6)$$

It is direct to see that the corresponding stationary zero curvature Eq. (1.2) yields the initial conditions:

$$a_x^{[0]} = 0, \quad b^{[0]} = c^{[0]} = e^{[0]} = g^{[0]} = 0, \quad f_x^{[0]} = 0, \quad (2.7)$$

and the recursion relations for defining the Laurent series solution:

$$\begin{cases} b^{[n+1]} = \frac{1}{\alpha} (b_x^{[n]} + 2\delta_1 a^{[n]} u_1 + 2\delta_2 f^{[n]} u_2), \\ c^{[n+1]} = -\frac{1}{\alpha} (c_x^{[n]} - 2\delta_1 a^{[n]} u_3 + 2\delta_2 f^{[n]} u_4), \end{cases} \quad (2.8)$$

$$\begin{cases} e^{[n+1]} = \frac{1}{\alpha} (e_x^{[n]} - 2\delta_1 f^{[n]} u_1 + 2\delta_1 a^{[n]} u_2), \\ g^{[n+1]} = -\frac{1}{\alpha} (g_x^{[n]} - 2\delta_1 f^{[n]} u_3 - 2\delta_1 a^{[n]} u_4), \end{cases} \quad (2.9)$$

$$\begin{cases} a_x^{[n+1]} = \delta_1 c^{[n+1]} u_1 + \delta_2 g^{[n+1]} u_2 - \delta_1 b^{[n+1]} u_3 - \delta_2 e^{[n+1]} u_4, \\ f_x^{[n+1]} = \delta_1 g^{[n+1]} u_1 - \delta_1 c^{[n+1]} u_2 + \delta_1 e^{[n+1]} u_3 - \delta_1 b^{[n+1]} u_4, \end{cases} \quad (2.10)$$

where $n \geq 0$. To present the Laurent series solution concretely, we take the initial data,

$$a^{[0]} = \frac{1}{2}\beta, \quad f^{[0]} = \frac{1}{2}\gamma, \quad (2.11)$$

where β and γ are two arbitrary constants, and choose the constants of integration to be zero,

$$a^{[n]}|_{u=0} = 0, \quad f^{[n]}|_{u=0} = 0, \quad n \geq 1. \quad (2.12)$$

In this way, one can work out that

$$\begin{cases} b^{[1]} = \frac{1}{\alpha} (\delta_1 \beta u_1 + \delta_2 \gamma u_2), & c^{[1]} = \frac{1}{\alpha} (\delta_1 \beta u_3 - \delta_2 \gamma u_4), \\ e^{[1]} = -\frac{1}{\alpha} \delta_1 (\gamma u_1 - \beta u_2), & g^{[1]} = \frac{1}{\alpha} \delta_1 (\gamma u_3 + \beta u_4), \\ a^{[1]} = f^{[1]} = 0; \\ b^{[2]} = \frac{1}{\alpha^2} (\delta_1 \beta u_{1,x} + \delta_2 \gamma u_{2,x}), & c^{[2]} = -\frac{1}{\alpha^2} (\delta_1 \beta u_{3,x} - \delta_2 \gamma u_{4,x}), \\ e^{[2]} = -\frac{1}{\alpha^2} \delta_1 (\gamma u_{1,x} - \beta u_{2,x}), & g^{[2]} = -\frac{1}{\alpha^2} \delta_1 (\gamma u_{3,x} + \beta u_{4,x}), \\ a^{[2]} = -\frac{1}{\alpha^2} \delta_1 [(\delta_1 \gamma u_3 - \delta_2 \gamma u_4) u_1 + \delta_2 (\gamma u_3 + \beta u_4) u_2], \\ f^{[2]} = -\frac{1}{\alpha^2} \delta_1 [\delta_1 (\gamma u_3 + \beta u_4) u_1 - (\delta_1 \beta u_3 - \delta_2 \gamma u_4) u_2]; \\ b^{[3]} = \frac{1}{\alpha^3} [\delta_1 \beta u_{1,xx} + \delta_2 \gamma u_{2,xx} - 2\delta_1 (\delta_1 \beta u_3 - \delta_2 \gamma u_4) (\delta_1 u_1^2 - \delta_2 u_2^2) - 4\delta_1^2 \delta_2 (\gamma u_3 + \beta u_4) u_1 u_2], \\ c^{[3]} = \frac{1}{\alpha^3} [\delta_1 \beta u_{3,xx} - \delta_2 \gamma u_{4,xx} - 2\delta_1 (\delta_1 \beta u_1 + \delta_2 \gamma u_2) (\delta_1 u_3^2 - \delta_2 u_4^2) + 4\delta_1^2 \delta_2 (\gamma u_1 - \beta u_2) u_3 u_4], \\ e^{[3]} = \frac{1}{\alpha^3} [-\delta_1 \gamma u_{1,xx} + \delta_1 \beta u_{2,xx} + 2\delta_1^2 (\gamma u_3 + \beta u_4) (\delta_1 u_1^2 - \delta_2 u_2^2) - 4\delta_1^2 (\delta_1 \beta u_3 - \delta_2 \gamma u_4) u_1 u_2], \\ g^{[3]} = \frac{1}{\alpha^3} [\delta_1 \gamma u_{3,xx} + \delta_1 \beta u_{4,xx} - 2\delta_1^2 (\gamma u_1 - \beta u_2) (\delta_1 u_3^2 - \delta_2 u_4^2) - 4\delta_1^2 (\delta_1 \beta u_1 + \delta_2 \gamma u_2) u_3 u_4], \\ a^{[3]} = \frac{\delta_1}{\alpha^3} [-\delta_1 (\beta u_3 - \delta_2 \gamma u_4) u_{1,x} - \delta_2 (\gamma u_3 + \beta u_4) u_{2,x} + (\delta_1 \beta u_1 + \delta_2 \gamma u_2) u_{3,x} - \delta_2 (\gamma u_1 - \beta u_2) u_{4,x}], \\ f^{[3]} = \frac{\delta_1}{\alpha^3} [-\delta_1 (\gamma u_3 + \beta u_4) u_{1,x} + (\delta_1 \beta u_3 - \delta_2 \gamma u_4) u_{2,x} + \delta_1 (\gamma u_1 - \beta u_2) u_{3,x} + \delta_1 (\beta u_1 + \delta_2 \gamma u_2) u_{4,x}]; \end{cases}$$

and

$$\begin{cases} b^{[4]} = \frac{1}{\alpha^4} \{ \delta_1 \beta u_{1,xxx} + \delta_2 \gamma u_{2,xxx} - 6\delta_1^2 [u_1 (\delta_1 \beta u_3 - \delta_2 \gamma u_4) + \delta_2 u_2 (\gamma u_3 + \beta u_4)] u_{1,x} \\ \quad - 6\delta_1 \delta_2 [\delta_1 u_1 (\gamma u_3 + \beta u_4) - u_2 (\delta_1 \beta u_3 - \delta_2 \gamma u_4)] u_{2,x} \}, \\ c^{[4]} = -\frac{1}{\alpha^4} \{ \delta_1 \beta u_{3,xxx} - \delta_2 \gamma u_{4,xxx} - 6\delta_1^2 [u_1 (\delta_1 \beta u_3 - \delta_2 \gamma u_4) + \delta_2 u_2 (\gamma u_3 + \beta u_4)] u_{3,x} \\ \quad + 6\delta_1 \delta_2 [\delta_1 u_1 (\gamma u_3 + \beta u_4) - u_2 (\delta_1 \beta u_3 - \delta_2 \gamma u_4)] u_{4,x} \}, \\ e^{[4]} = \frac{1}{\alpha^4} \{ -\delta_1 \gamma u_{1,xxx} + \delta_1 \beta u_{2,xxx} + 6\delta_1^2 [\delta_1 u_1 (\gamma u_3 + \beta u_4) - u_2 (\delta_1 \beta u_3 - \delta_2 \gamma u_4)] u_{1,x} \\ \quad - 6\delta_1^2 [u_1 (\delta_1 \beta u_3 - \delta_2 \gamma u_4) + \delta_2 u_2 (\gamma u_3 + \beta u_4)] u_{2,x} \}, \\ g^{[4]} = -\frac{1}{\alpha^4} \{ \delta_1 \gamma u_{3,xxx} + \delta_1 \beta u_{4,xxx} - 6\delta_1^2 [\delta_1 u_1 (\gamma u_3 + \beta u_4) - u_2 (\delta_1 \beta u_3 - \delta_2 \gamma u_4)] u_{3,x} \\ \quad - 6\delta_1^2 [u_1 (\delta_1 \beta u_3 - \delta_2 \gamma u_4) + \delta_2 u_2 (\gamma u_3 + \beta u_4)] u_{4,x} \}, \\ a^{[4]} = \frac{\delta_1}{\alpha^4} [-\delta_1 (\beta u_3 - \delta_2 \gamma u_4) u_{1,xx} - \delta_2 (\gamma u_3 + \beta u_4) u_{2,xx} - (\delta_1 \beta u_1 + \delta_2 \gamma u_2) u_{3,xx} \\ \quad + \delta_2 (\gamma u_1 - \beta u_2) u_{4,xx} + (\delta_1 \beta u_{3,x} - \delta_2 \gamma u_{4,x}) u_{1,x} + \delta_2 (\gamma u_{3,x} + \beta u_{4,x}) u_{2,x} \\ \quad + 3\delta_1^2 (\delta_1 \beta u_3^2 - 2\delta_2 \gamma u_3 u_4 - \delta_2 \beta u_4^2) u_1^2 + 6\delta_1 \delta_2 (\delta_1 \gamma u_3^2 + 2\delta_1 \beta u_3 u_4 - \delta_2 \gamma u_4^2) u_1 u_2 \\ \quad - 3\delta_1 \delta_2 (\delta_1 \beta u_3^2 - 2\delta_2 \gamma u_3 u_4 - \delta_2 \beta u_4^2) u_2^2], \end{cases}$$

$$\begin{aligned}
f^{[4]} = & \frac{\delta_1}{\alpha^4} [-\delta_1(\gamma u_3 + \beta u_4)u_{1,xx} + (\delta_1\beta u_3 - \delta_2\gamma u_4)u_{2,xx} - \delta_1(\gamma u_1 - \beta u_2)u_{3,xx} \\
& - (\delta_1\beta u_1 + \delta_2\gamma u_2)u_{4,xx} + \delta_1(\gamma u_{3,x} + \beta u_{4,x})u_{1,x} - (\delta_1\beta u_{3,x} - \delta_2\gamma u_{4,x})u_{2,x} \\
& + 3\delta_1^2(\delta_1\gamma u_3^2 + 2\delta_1\beta u_3u_4 - \delta_2\gamma u_4^2)u_1^2 - 6\delta_1^2(\delta_1\beta u_3^2 - 2\delta_2\gamma u_3u_4 - \delta_2\beta u_4^2)u_1u_2 \\
& - 3\delta_1\delta_2(\delta_1\gamma u_3^2 + 2\delta_1\beta u_3u_4 - \delta_2\gamma u_4^2)u_2^2].
\end{aligned}$$

Upon taking these results into consideration, we can impose $\Delta_r = 0$, $m \geq 0$, to formulate

$$\varphi_{t_m} = \mathcal{N}^{[m]} \varphi = \mathcal{N}^{[m]}(u, \lambda) \varphi, \quad \mathcal{N}^{[m]} = (\lambda^m Y)_+ = \sum_{n=0}^m \lambda^n Y^{[m-n]}, \quad m \geq 0, \quad (2.13)$$

which are the temporal matrix eigenvalue problems. The corresponding zero curvature equations in (1.5) generate a soliton hierarchy with four potentials:

$$u_{t_m} = X^{[m]} = X^{[m]}(u) = (\alpha b^{[m+1]}, \alpha e^{[m+1]}, -\alpha c^{[m+1]}, -\alpha g^{[m+1]})^T, \quad m \geq 0. \quad (2.14)$$

As illustrative examples, this hierarchy gives us the model of combined integrable nonlinear Schrödinger equations:

$$u_{1,t_2} = \alpha b^{[3]}, \quad u_{2,t_2} = \alpha e^{[3]}, \quad u_{3,t_2} = -\alpha c^{[3]}, \quad u_{4,t_2} = -\alpha g^{[3]}, \quad (2.15)$$

and the model of combined integrable modified Korteweg–de Vries equations:

$$u_{1,t_3} = \alpha b^{[4]}, \quad u_{2,t_3} = \alpha e^{[4]}, \quad u_{3,t_3} = -\alpha c^{[4]}, \quad u_{4,t_3} = -\alpha g^{[4]}, \quad (2.16)$$

where $b^{[m]}, c^{[m]}, e^{[m]}$ and $g^{[m]}$, $m = 3, 4$, are presented previously. The two systems are typical coupled integrable models, which extend the category of coupled integrable models of nonlinear Schrödinger equations and modified Korteweg–de Vries equations (see, e.g., [23,27,28]). One particular characteristic is that every equation contains two derivative terms of the highest order, and so, we call them combined models.

In the hierarchy (2.14), three special cases with $\delta_2 = 0$, $\beta = 0$ and $\gamma = 0$ are of interest. The first case produces novel integrable couplings of the AKNS hierarchy, which are not of perturbation type. The other two cases present reduced hierarchies of uncombined integrable models.

3. Bi-Hamiltonian formulations

Let $\delta_1\delta_2 \neq 0$ now. To establish bi-Hamiltonian formulations [29,30] and show the Liouville integrability for the soliton hierarchy (2.14), we can apply the so-called trace identity (1.7). We can readily work out

$$\text{tr}\left(Y \frac{\partial \mathcal{M}}{\partial \lambda}\right) = 2\alpha\delta_1 a, \quad \text{tr}\left(Y \frac{\partial \mathcal{M}}{\partial u}\right) = (2\delta_1^2 c, 2\delta_1\delta_2 g, 2\delta_1^2 b, 2\delta_1\delta_2 e)^T, \quad (3.1)$$

and consequently, the trace identity produces

$$\frac{\delta}{\delta u} \int \lambda^{-(n+1)} \alpha a^{[n+1]} dx = \lambda^{-\kappa} \frac{\partial}{\partial \lambda} \lambda^{\kappa-n} (\delta_1 c^{[n]}, \delta_2 g^{[n]}, \delta_1 b^{[n]}, \delta_2 e^{[n]})^T, \quad n \geq 0. \quad (3.2)$$

Checking with $n = 2$ leads to $\kappa = 0$, and accordingly, we arrive at

$$\frac{\delta}{\delta u} \mathcal{H}^{[n]} = (\delta_1 c^{[n+1]}, \delta_2 g^{[n+1]}, \delta_1 b^{[n+1]}, \delta_2 e^{[n+1]})^T, \quad n \geq 0, \quad (3.3)$$

as soon as we take the Hamiltonian functionals as

$$\mathcal{H}^{[n]} = - \int \frac{\alpha a^{[n+2]}}{n+1} dx, \quad n \geq 0. \quad (3.4)$$

This allows us to propose a Hamiltonian formulation for the hierarchy (2.14):

$$u_{t_m} = X^{[m]} = J_1 \frac{\delta \mathcal{H}^{[m]}}{\delta u}, \quad J_1 = \begin{bmatrix} 0 & M_1 \\ -M_1 & 0 \end{bmatrix}, \quad M_1 = \begin{bmatrix} \frac{\alpha}{\delta_1} & 0 \\ 0 & \frac{\alpha}{\delta_2} \end{bmatrix}, \quad m \geq 0, \quad (3.5)$$

The Hamiltonian theory tells that there is an interrelation $S = J_1 \frac{\delta \mathcal{H}}{\delta u}$ between a symmetry S and a conserved functional $\mathcal{H}$ of the same model.

These vector fields $X^{[n]}$ commute with each other:

$$[[X^{[n_1]}, X^{[n_2]}]] = X^{[n_1]'}(u)[X^{[n_2]}] - X^{[n_2]'}(u)[X^{[n_1]}] = 0, \quad n_1, n_2 \geq 0. \quad (3.6)$$

This property can be obtained from an abelian algebra of Lax operators:

$$[[\mathcal{N}^{[n_1]}, \mathcal{N}^{[n_2]}]] = \mathcal{N}^{[n_1]'}(u)[X^{[n_2]}] - \mathcal{N}^{[n_2]'}(u)[X^{[n_1]}] + [\mathcal{N}^{[n_1]}, \mathcal{N}^{[n_2]}] = 0, \quad n_1, n_2 \geq 0,$$

which can directly be proved by analyzing the relation between the isospectral zero curvature equations (see [31] for the detailed proof).

Furthermore, based on the recursion relation $X^{[m+1]} = \Phi X^{[m]}$, we can compute a hereditary recursion operator $\Phi = (\Phi_{jk})_{4 \times 4}$ [30] for the hierarchy (2.14) as follows:

$$\begin{cases} \Phi_{11} = \frac{2}{\alpha}(\frac{1}{2}\partial_x - \delta_1 u_1 \partial^{-1} u_3 - \delta_2 u_2 \partial^{-1} u_4), & \Phi_{12} = \frac{2}{\alpha}(-\delta_2 u_1 \partial^{-1} u_4 + \delta_2 u_2 \partial^{-1} u_3), \\ \Phi_{13} = \frac{2}{\alpha}(-\delta_1 u_1 \partial^{-1} u_1 + \delta_2 u_2 \partial^{-1} u_2), & \Phi_{14} = \frac{2}{\alpha}(-\delta_2 u_1 \partial^{-1} u_2 - \delta_2 u_2 \partial^{-1} u_1); \end{cases} \quad (3.7)$$

$$\begin{cases} \Phi_{21} = \frac{2}{\alpha}(\delta_1 u_1 \partial^{-1} u_4 - \delta_1 u_2 \partial^{-1} u_3), & \Phi_{22} = \frac{2}{\alpha}(\frac{1}{2}\partial_x - \delta_1 u_1 \partial^{-1} u_3 - \delta_2 u_2 \partial^{-1} u_4), \\ \Phi_{23} = \frac{2}{\alpha}(-\delta_1 u_1 \partial^{-1} u_2 - \delta_1 u_2 \partial^{-1} u_1), & \Phi_{24} = \frac{2}{\alpha}(\delta_1 u_1 \partial^{-1} u_1 - \delta_2 u_2 \partial^{-1} u_2); \end{cases} \quad (3.8)$$

$$\begin{cases} \Phi_{31} = \frac{2}{\alpha}(\delta_1 u_3 \partial^{-1} u_3 - \delta_2 u_4 \partial^{-1} u_4), & \Phi_{32} = \frac{2}{\alpha}(\delta_2 u_3 \partial^{-1} u_4 + \delta_2 u_4 \partial^{-1} u_3), \\ \Phi_{33} = \frac{2}{\alpha}(-\frac{1}{2}\partial_x + \delta_1 u_3 \partial^{-1} u_1 + \delta_2 u_4 \partial^{-1} u_2), & \Phi_{34} = \frac{2}{\alpha}(\delta_2 u_3 \partial^{-1} u_2 - \delta_2 u_4 \partial^{-1} u_1); \end{cases} \quad (3.9)$$

$$\begin{cases} \Phi_{41} = \frac{2}{\alpha}(\delta_1 u_3 \partial^{-1} u_4 + \delta_1 u_4 \partial^{-1} u_3), & \Phi_{42} = \frac{2}{\alpha}(-\delta_1 u_3 \partial^{-1} u_3 + \delta_2 u_4 \partial^{-1} u_4), \\ \Phi_{43} = \frac{2}{\alpha}(-\delta_1 u_3 \partial^{-1} u_2 + \delta_1 u_4 \partial^{-1} u_1), & \Phi_{44} = \frac{2}{\alpha}(-\frac{1}{2}\partial_x + \delta_1 u_3 \partial^{-1} u_1 + \delta_2 u_4 \partial^{-1} u_2). \end{cases} \quad (3.10)$$

With some analysis, we can verify that J_1 and $J_2 = \Phi J_1$ constitute a Hamiltonian pair, and thus all equations, except the first one, in the hierarchy (2.14) possess a bi-Hamiltonian formulation [29]:

$$u_{t_m} = X^{[m]} = J_1 \frac{\delta \mathcal{H}^{[m]}}{\delta u} = J_2 \frac{\delta \mathcal{H}^{[m-1]}}{\delta u}, \quad m \geq 1. \quad (3.11)$$

We can further show that the associated Hamiltonian functionals commute with each other under each of the corresponding two Poisson brackets [6]:

$$\{\mathcal{H}^{[n_1]}, \mathcal{H}^{[n_2]}\}_{J_k} = \int \left(\frac{\delta \mathcal{H}^{[n_1]}}{\delta u} \right)^T J_k \frac{\delta \mathcal{H}^{[n_2]}}{\delta u} dx = 0, \quad k = 1, 2, \quad n_1, n_2 \geq 0. \quad (3.12)$$

To summarize, all models in the hierarchy (2.14) are Liouville integrable and possess infinitely many commuting symmetries $\{X^{[n]}\}_{n=0}^{\infty}$ and conserved functionals $\{\mathcal{H}^{[n]}\}_{n=0}^{\infty}$. Particularly, the two systems in (2.15) and (2.16) provide concrete examples of nonlinear combined Liouville integrable Hamiltonian models.

4. Concluding remarks

From a specific special matrix eigenvalue problem, a Liouville four-component integrable hierarchy with bi-Hamiltonian formulations has been constructed. The success lies in the use of a particular Laurent series solution of the corresponding stationary zero curvature equation.

We remark that the case of $\delta_2 = 0$ generates integrable couplings and the variational identity is required to furnish a Hamiltonian formulation (see, e.g., [5] for details) and the case of $\delta_1 = 1$ reduces to a spectral eigenvalue problem discussed earlier [32]. It should be of particular interest to study mathematical formulations of solitons to multi-component integrable models. Various effective approaches to solitons contain the Riemann–Hilbert technique [33], the determinant approach [34], the Darboux transformation [35–37] and the Zakharov–Shabat dressing method [38]. Besides soliton solutions, lump, kink, breather and rogue wave solutions and their interaction solutions (see, e.g., [39–45]), are also interesting, which can often be computed from solitons. In addition, taking nonlocal group reductions of matrix eigenvalue problems, we can formulate nonlocal reduced integrable models and their solitons (see, e.g., [46–48]).

Integrable models are of great interest. They significantly advance our understanding of solitons and other coherent structures in complex nonlinear problems (see, e.g., [49]). They provide practical insights into the dynamic behavior of physical systems and have close connections to diverse areas of mathematics, including algebraic geometry, Hamiltonian dynamics, and the theory of Lie groups and algebras.

Data availability

No data was used for the research described in the article.

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